

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

_____ show ip route ospf _____

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? _____ S0/0/0 _____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? _____ 129 _____

Does R2 show up as an OSPF neighbor on R1? _____ Yes _____

Does R2 show up as an OSPF neighbor on R3? _____ No _____

What does this information tell you?

_____ All traffic to the 192.168.2.0/24 network from R3 will be routed through R1. The S0/0/1 interface on R2 is still configured as a passive interface, so OSPF routing information is not being advertised across this interface. _____

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

_____ R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1 _____

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? _____ S0/0/1 _____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

_____ 65. One T1 (1.544 Mb/s) serial link (at a cost of 64) plus the R2 Gigabit 0/0 LAN link (at a cost of 1) _____

Is R2 listed as an OSPF neighbor to R3? __Yes__

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
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Why would you want to change the OSPF default reference-bandwidth?

____To accommodate higher-speed links (e.g., Gigabit Ethernet) so that the cost calculation yields distinct values for different speeds, preventing multiple interfaces from having the same cost and allowing more accurate path selection._____

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

____192.168.3.0/24: R1 S0/0/1 + R3 G0/0 (781+1=782). 192.168.23.0/30: R1 S0/0/1 and R3 S0/0/1 (781+64=845)._____

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

____1,562. Each serial link now has a cost of 781, and the route to the 192.168.23.0/24 network travels over two serial links. 781 + 781 = 1,562._____

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

____OSPF will choose the route with the least accumulated cost. The route with the lowest accumulated cost is: R1-S0/0/0 + R2-S0/0/1 + R3-G0/0, or 781 + 781 + 1 = 1,563. _____

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

___ Router ID uniquely identifies a router in the OSPF domain and is used in DR/BDR elections and LSA advertisements. If not manually set, OSPF dynamically selects the highest IP address on a loopback or physical interface, which can change if interfaces are added or removed, causing network instability and unnecessary adjacencies. Manual assignment ensures predictability and stability. _____

2. Why is the DR/BDR election process not a concern in this lab?

___ All links in this lab are point-to-point serial connections. DR/BDR election only occurs on broadcast multi-access networks (e.g., Ethernet). On point-to-point links, no election takes place, simplifying OSPF operation. _____

3. Why would you want to set an OSPF interface to passive?

___ To prevent OSPF from sending hello packets on that interface, which reduces unnecessary traffic and enhances security on networks that do not have any other OSPF neighbors (e.g., a LAN with only end hosts). The network connected to the passive interface is still advertised to other OSPF routers. _____

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

___ The use of areas. OSPF divides a network into areas, with Area 0 (backbone) as the core. This limits the flooding of LSAs within an area, allows route summarization, and reduces routing table size, making OSPF highly scalable. _____

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

___ network 10.0.0.0 0.255.255.255 area 0 _____

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

___ The EIGRP route will be used because it has the lowest administrative distance (EIGRP internal AD = 90, OSPF = 110, RIP = 120). The metric is only compared among routes from the same protocol. _____

